



Indian Institute of Technology, Kanpur
Computer Science and Engineering

Quiz 3

CS670: Cryptographic Techniques for Privacy Preservation

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03/11/2025

Name: _____

Roll Number: _____

This quiz contains 6 pages (including this cover page) and 8 questions. Total of marks is 25.

Distribution of Marks

Question	Points	Score
1	2	
2	3	
3	3	
4	2	
5	4	
6	4	
7	2	
8	5	
Total:	25	

1. Describe in one sentence what the following properties of commitments schemes are:

(a) (1 point) Computational Hiding

Solution: By looking at the commitment the receiver cannot make out *what* was committed to.

(b) (1 point) Computational Binding

Solution: Once the receiver has committed to a value, they cannot change it.

2. (3 points) Describe a simple *zero knowledge proof* to prove that a prover knows a 3-coloring of a graph. Why do we need repeat the experiment?

Solution: The prover colors the graph and then commits to all the colors. The verifier then picks a random edge and asks the prover to decommit the colors of the vertices corresponding to the edge. If the vertices are colored differently, the verifier accepts the proof. The verifier needs to repeat the experiment in order to reduce the probability of the prover cheating.

3. (3 points) Recall MPC-in-the-Head Protocol. Describe in short why we cannot use a 1-private protocol to do the MPC-in-the-head where views of two parties are opened?

Solution: In MPC-in-the-Head, the verifier asks to open the internal views of two out of the three simulated parties. A 1-private protocol only guarantees privacy as long as *no* party colludes with another. Revealing the views of two parties is equivalent to those two parties colluding, which breaks the 1-privacy guarantee and leaks the secret.

4. (2 points) What change must be made to allow for 1-private protocols to be run in the head?

Solution: To use a 1-private protocol in MPC-in-the-Head, we must avoid revealing the full internal views of two parties. Instead, we open the view of only *one* party together with the messages that this party receives (i.e., the incoming communication edges). This prevents the reconstruction of all shares and preserves 1-privacy.

5. (4 points) Describe the basic structure of the Schnorr protocol (the Σ protocol).

Solution: Schnorr Protocol is a three-move protocol. In the first move the prover sends a commitment. In the second move, the verifier sends a challenge to the prover. Finally in the third move, prover sends a response. The verifier accepts if it is a valid response.

6. (4 points) What is the special soundness property with respect to Zero Knowledge Proofs. How does the Schnorr Protocol satisfy the special soundness property?

Solution: Special Soundness: A Σ -protocol has the *special soundness* property if, from two accepting transcripts that share the same first message (commitment) but have different verifier challenges, one can efficiently extract the witness. In other words, if a prover can answer two different challenges for the same initial commitment, then the prover must “know” the witness.

Schnorr Protocol: In the Schnorr identification protocol, the prover wants to prove knowledge of a secret w such that $h = g^w \bmod p$.

The messages are:

$$\text{Commit: } a = g^r, \quad \text{Challenge: } c, \quad \text{Response: } z = r + cw \pmod{q}.$$

Suppose we have two accepting transcripts with the same a but different challenges c and c' :

$$(a, c, z) \quad \text{and} \quad (a, c', z').$$

Then,

$$z = r + cw \quad \text{and} \quad z' = r + c'w.$$

Subtracting gives:

$$z - z' = (c - c')w \pmod{q},$$

so

$$w = (z - z')(c - c')^{-1} \pmod{q}.$$

Thus, from two valid transcripts, the witness w can be efficiently recovered. Therefore, the Schnorr protocol satisfies the special soundness property.

7. (2 points) What is the technique that is used to prove *zero knowledge*?

Solution: Show the existence of a simulator

8. (5 points) A **Hamiltonian cycle** in a graph $G = (V, E)$ is a cycle that visits every vertex in V exactly once and returns to the starting vertex. Formally, a Hamiltonian cycle is a sequence of distinct vertices

$$(v_1, v_2, \dots, v_n)$$

such that

$$(v_i, v_{i+1}) \in E \quad \text{for all } i = 1, 2, \dots, n-1,$$

and

$$(v_n, v_1) \in E.$$

Determining whether such a cycle exists is an **NP**-complete problem.

The goal is for the prover P to convince the verifier V that the graph G contains a Hamiltonian cycle, without revealing the cycle itself.

Solution:

1. The prover P knows a Hamiltonian cycle C in G . P chooses a random permutation π on the vertices and constructs the isomorphic graph $H = \pi(G)$. P commits to the adjacency matrix (or adjacency lists) of H and sends the commitments to the verifier V .
2. V sends a random challenge bit $b \in \{0, 1\}$.
3. If $b = 0$, P reveals π and opens *all* commitments, allowing V to verify that H is isomorphic to G .
4. If $b = 1$, P reveals only the cycle $\pi(C)$ and opens the commitments corresponding to the edges of this cycle. V verifies that these edges indeed form a Hamiltonian cycle in H , and learns nothing else.

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